Comments of
The Health Record Banking Alliance
In response to

Department of Health and Human Services
Centers for Medicare and Medicaid Services
Request for Information – Health Technology Ecosystem
Reference File Code CMS–0042–NC

Submitted June 14, 2025 via http://www.regulations.gov

Our country now has technology using FHIR to integrate health information around each patient. Integrating encounter records around the patient is far more efficient than the present, historically established non-system that stores patient records in scattered pieces among various hospital and medical office EHR databases.

FHIR technology thus can lead to a new system design featuring lifetime health records each patient owns and controls, and are exchangeable interoperably and securely. These are explicit 21st Century Cures Act goals; they are mandatory, and they have effectively been ignored over the last four years by ASTP/ONC in its TEFCA implementation.

To accomplish what the Cures Act demands, CMS and ONC have the opportunity and obligation to adopt a national health data architecture based on private sector Cures Act Repository/Health Data Banks ("Cures Act Repositories" or "HDBs"). Accounts in those repositories (similar to financial checking accounts) will hold patients' lifetime, "longitudinal," problem-oriented health records. Patients, clinicians, and researchers whom patients want to support all will benefit, as we explain below.

A national health data architecture designed this way will dramatically enhance the security, efficiency, and capacity of health information flows in the United States. It will fulfill Cures Act specifications.

CMS should note that those mandatory specifications will go unfulfilled unless ASTP/ONC's current plans for implementing the Cures Act's TEFCA mandate are redirected to the HDB/Cures Act Repository-based architecture and system design we propose here.

These comments, filed by the Health Record Banking Alliance ("HRBA")[1] explain how CMS, in concert with ASTP/ONC and other agencies, can accomplish this re-working of our national health data architecture. All of the technology to make it work exists. FHIR solves data format issues for storage and exchange, and Trust Over IP provides the tools to handle encryption-decryption, messaging, and key management; and multiple NIST IAL2 certified identity management companies provide verifiable identity management for patients. Once the HDB/Cures Act Repository-based architecture is in place and its adoption begins, the system

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[1] The Health Record Banking Alliance, P.O. Box 6580, Falls Church, Virginia 22040, is recognized as a business league by the Internal Revenue Service under Section 501(c)(6) of the Internal Revenue Code.

design can expand to create new efficiencies for high-capacity digital exchange of health data among health industry institutions. The systems design thus can accommodate institutional needs for secure exchange of interoperable data for TPO (Treatment, Payment, and Health Care Operations) that are necessary to complement HIPAA standard transactions for billing and related functions.

However, if CMS in concert with other agencies cannot dissuade ASTP/ONC from its current flawed concept for implementing TEFCA, then ASTP/ONC and CMS will continue pouring resources into an inefficient, insecure, and otherwise unworkable version of TEFCA. Not incidentally, that concept of TEFCA contravenes the Cures Act's overarching policy goal of patients' control of their health records. It also patently violates the Act's explicit high-level engineering specifications that compel a new health data architecture for the nation.

The result would be another lost decade in the heretofore elusive quest for health data exchangeability, interoperability, security, and all the benefits those attributes will bring – the goal set out by CMS in the instant RFI:

> . . . to implement certain policies to enhance the amount of health data available through digital channels and give patients secure, electronic access to their personal health information. These policies are the building blocks of a digital ecosystem in which patients can view, manage, utilize, and share their health information through digital applications (apps) and other modern solutions. This digital ecosystem could ultimately expand access to personalized health guidance for patients to improve prevention and chronic disease management. CMS, *Request of Information; Health Technology Ecosystem,* 90 Fed. Reg. 21034, 21035 (May 16, 2025).

The following analysis addresses questions posed in the RFI by answering in terms of an overall systems analysis of the architectural challenge CMS is presenting. We have not sought to answer each designated question separately.

What Is a Cures Act Repository/Health Data Bank and What National Ecosystem Can Result from Adoption of CAR/HDBs?

Appendix 1 is a graphic with accompanying text explanation of a Health Data Bank and the highly efficient, expanded data flows it enables.

Appendix 2 is a graphic presentation of the health data ecosystem that can result once a HDB/Cures Act Repository-based architecture is implemented through redirection of ASTP/ONC and CMS policy to comport with the Cures Act specifications described in the previous section.

These two appendices will help orient readers to the discussion that follows.

Cures Act Mandatory Specifications for A Patient-Owned and -Controlled Lifetime Health Record

Promises were made but not kept in the 21st Century Cures Act. A key Congressional goal – a fundamental Cures Act promise – is assuring patients access to their electronic health

information "*in a single, longitudinal format that is easy to understand, secure, and may be updated automatically.*" (HITECH new §3009(e)(2)(A) as added by Cures Act §4006; emphasis added.)

The 21st Century Cures Act was signed into law on December 13, 2016. Yet to date, years later, no executive action at HHS or ASTP/ONC realistically implements the explicit design specifications that the Cures Act prescribes. Questions arise.

First, "Where is the patient?" Second: "Where is the physician?"

And a third question, fundamental to national security: Why is ASTP/ONC implementing a national health data infrastructure based on Qualified Health Information Networks (QHINs) that will be inefficient, inordinately complex and troublesome in terms of connectivity and operating protocols, difficult for patients and physicians to use, *and* fundamentally insecure and inherently vulnerable to continuous attack by hostile actors?

Contrast these difficulties with what is envisioned in the Cures Act, set out in detail below. Text in blue underscores TEFCA design imperatives for data integration around each patient. Systems designed to meet these specifications must, realistically, be integral to engineering plans from the beginning of implementation. They cannot effectively be layered on later. This is one aspect of the reality that condemns ASTP/ONC's current TEFCA design to failure.

The Cures Act says:

- EHR data exchange *must not* be implemented by ASTP/ONC in ways that *"are likely to substantially increase the complexity or burden"* of access to, or exchange of, health information. (HITECH new section 3022(a)(2)(B) as added by Cures Act section 4004; emphasis added.) This provision, which necessarily imposes a specific requirement for nationwide standardized exchange, cannot be satisfied by ONC's TEFCA architecture.

- EHR data must be exchangeable "*without special effort"* on the part of users. (Patients and physicians are among "users" under the Cures Act.) (HITECH as amended, new §3000(10)(A), as added by Cures Act section 4003(a); emphasis added.)

- EHR data exchange must provide *the patient or an authorized designee* with a *complete copy* of his or her health information from an electronic record *in a computable format*. (HITECH new §3000(10)(B) as added by Cures Act section 4003; emphasis added.)

- EHR systems must be capable of exchanging health data with "clinician-led clinical data registries" that are "designed to collect detailed, standardized data . . . for medical procedures, services, or therapies for particular diseases, conditions or exposures," and that meet data quality standards including "using standardized data elements . . . to verify the completeness and validity of those data." (Cures Act §4005(a),(b).)

- ONC is specifically empowered using system certification criteria to support offering "patient access to their electronic health information *in a single, longitudinal format that is easy to understand, secure, and may be updated automatically*" and supports patients' ability to add "patient-reported information" electronically as well as patients' access (at

each patient's option) for research purposes. (HITECH new section 3009(e)(2)(A) as added by Cures Act section 4006; emphasis added.)

A notable feature of ASTP/ONC's QHIN TEFCA architecture is how thoroughly it disregards these Cures Act engineering design specifications. ASTP/ONC thus fails to meet Congress's Cures Act goal of giving patients a *computable, complete, easy to understand, automatically updated, lifetime health record in a secure, easy to understand format* that patients can obtain *without special effort* and where the data exchange with Electronic Medical Record (EMR) systems in hospitals and physician offices can be accomplished *without substantially increase[ing] the complexity or burden* of accessing and exchanging data with those institutional systems.

Among the fallout from ASTP/ONC's failure to follow Cures Act specifications are the following basic system design concerns:

- **Patient Access.** Patient access to their medical data in institutional EMR systems through ASTP/ONC's QHIN architecture is, as a practical matter, not possible now and not likely to be possible in any reasonable view of the future.

- **Patient Complexity.** ASTP/ONC's QHIN architecture – if it ever were to be successfully engineered – will be so complex and burdensome for patients to navigate that they will not use it.

- **Provider Access.** Similarly, providers will not use ASTP/ONC's QHIN architecture because of the time-consuming and costly burdens of querying QHINs to assemble a patient's scattered records, and then going through elaborate procedures to assure that the records they are seeking and receiving apply to the patient they are treating, and not to one or more other patients.

- **Data Quality and Reliability.** Cobbling together patient records on the fly at a point of care, which is the operational mode that ASTP/ONC's QHIN architecture envisions, is an invitation to data quality problems. In contrast, compiling data as we propose in Cures Act Repositories affords consistent processes. There will be Cures Act Repository software systems to analyze each patient's data; identify, flag, and resolve conflicts in data; and organize the data in problem-oriented records. These features will aid physicians and other clinicians at each point of care that a patient visits.

- **Patient Identification.** As we explain in Appendix 1, the description of a representative Cures Act Repository, the repository architecture will significantly mitigate patient matching problems for patients with repository accounts (akin to customer identification with financial institution accounts). In contrast, ASTP/ONC's QHIN architecture will be plagued by them.

- **Clinical Data to Support Medical Research.** Also as explained in Appendix 1, Cures Act repository architecture will be designed so patients can contribute their medical data to support research they care about. Data flows in Cures Act Repository architecture facilitate patients' sending all or part of their data, whether identified or anonymized, to researchers whose work they support.

Medical research today requires hundreds of thousands, if not millions, of vetted patient records. With that volume of records, advanced computational methods, including those using artificial intelligence (AI), can be employed to analyze vast troves of data. Many patients will want to support such research. They will learn about opportunities of interest through notification systems that Cures Act Repositories will establish. Researchers will use those notification systems to publicize their work and to request patient data they need to support their research. (Note that ASTP/ONC's QHIN architecture is so complex, operationally burdensome, and fragmented that it will stand as a barrier to patients' support of medical research.)

- **Scalability.** ASTP/ONC's QHIN architecture cannot support the continuing and massive data loads its operations envision. The data loads come from continuous "scattered system queries" relating to patients presenting at points of care. There will then be responsive system messages seeking to clarify and verify patient identity. Those message exchanges will be followed, in theory, by QHIN data base responses for fully identified patients. Those responses will be sent, supposedly immediately, to requesting clinicians at points of care. The network loads imposed by these processes explain why ASTP/ONC's QHIN architecture cannot operate at scale.

These problems compound a fundamental economic policy problem arising from ASTP/ONC's ignoring Cures Act requirements: the private sector is not encouraged, but rather is discouraged, by federal policy from pursuing application programming interfaces (APIs) to support patient-centered lifetime health records aggregated and stored in Cures Act Repositories.

Similarly, ASTP/ONC's policy divergence from the Cures Act discourages support for private sector development of software systems for Cures Act Repositories. Private sector technology design is important as well for secure point-to-point, FHIR-based exchange that will work hand-in-glove with Cures Act Repositories and enable their utility and proliferation. These private sector activities are integral to reforming health data exchange as the Cures Act envisions and demands.

TEFCA Architectural Implementation: Contrast Between Integration of Data Around Each Patient Versus ASTP/ONC's Current, Inefficient, Unworkable Model of Patients' Data Remaining Scattered Among Institutional EHR Databases

The following tables and annotations compare how an architecture for health information centered around patient-controlled accounts in Cures Act Repositories/HDBs will revolutionize information flows for care, provider convenience, and overall health system efficiency.

Tractable vs. Intractable, Centered vs. Scattered	**Centered ~ Cures Act Repository/HRB Model**	**Scattered ~ ASTP/ONC Current TEFCA Model**
Basics	Single designated and secure system which is: • Patient-controlled and provider neutral • Maintained by a trusted custodian organization Where the patient or their representative: • Maintains unique ID, account and address • Designates a single place to send/store their records, e.g., after each encounter	Patient data/records are managed across 10s and 100s of HINs and 1000s of systems/ devices, each of which maintains/manages: • Trusted software and storage • Accountability, authentication, authorization, consents, access control, audit mechanisms • Some fragment of the patient record
Query	Query is directed to a single designated Repository/HRB account for the identified patient	Broadcast query goes to 10s or 100s of HINs, then on to 1000s, 10,000s, 100,000s of systems/devices
Query response	• Response is immediate • All relevant and permitted records are immediately available • Responses are in searchable Problem-Oriented Health Records • You immediately know what you need to know	• Response may be nothing, trickle or deluge • Response content may vary with each query • Response may take minutes, hours or days • You don't know what you don't know • You don't know how long to wait
Confidentiality/ Authorization	Managed at a single point by each patient and/or patient representative/coach	Managed within a complex lattice of provider and HIN permissions plus patient consents
Real-time + Continuous	Sustained (24 x 7) support for individual health and healthcare – monitoring and guidance	Never, never land

A Center for the Individual – Making Data Integration Around the Patient a Reality

- No amount of information standards, policy frameworks or common agreements will answer these "Scatter Model" conundrums.
- The problem that needs solving is making the 'center' real: by creating the system that promotes and serves the individual.
- Providers then interact with that system as a routine part of care and as part of the duty of care, contributing to a persistent, coherent record of an individual's overall health and care, their journey and path forward.
- Providers make use of that record problem-oriented when needed – without recourse to interactions with other institutions.

- Providers continue to operate their own EHR/HIT systems for their own purposes – established as their business record – capturing the operational details of the care they provide and activities they perform.

A Center for the Individual – Effectiveness of the Centered Model	**Centered**	**Scattered**
Information Practices, Governance		
Rules and Logs • User authentication • Authorization, consents for information sharing • Access control • Minimum necessary and "need to know" • Audit events • Data exchange events	• Consistent, uniformly implemented • Can be controlled by patient, patient representative/coach Always: • Visible to patient • Visible to clinicians, as permitted	Across multiple source systems/devices, HINs/QHINs: • Disparate conventions, policies, governance • Controlled by multiple entities • Unevenly implemented Rules are: • Mostly invisible to patient • Mostly invisible to clinicians
Personal Parameters		
Personal • IDs, identity parameters • Demographics • Genetics • Health Data Use Directives • Patient Reported Outcomes • Social determinants of health • Insurance, coverage • Next of kin • Advance directives, health-related consents	• Consistent, uniformly implemented • Visible and can be managed by patient, patient representative/ coach	Across multiple source systems/devices, HINs/QHINs: • Disparate conventions, policies, governance • Controlled by multiple entities • Unevenly implemented

A Center for the Individual – Effectiveness of the Centered Model	Centered	Scattered
Clinical Findings		
Current and Historical • Encounters • Personal and family health history & physical findings • Vital signs • Assessments • Progress notes • Care records • Diagnostic results	Patient data in one place • Can be monitored directly by the patient • Can be annotated or corrected by the patient	Patient data in multiple disparate fragments across multiple source systems/ devices, HINs/QHINs Multiple portals and/or repeated "pulls" to gather "current" view
Coordinated Care		

Current and Historical
• Care Plan(s)
• Orders
• Therapies
• Consults
• Referrals
• Schedules
  • Appointments + followup
• Messages, interactions with caregivers
	Uniform, integrated
	• All patient data
	• All providers
	• All activities
	  • Past, present, future
	In real-time, patient can:
	• Monitor data and actions
	• Be prompted to take actions
	• Complete and document actions taken
		Patient data in multiple disparate fragments across multiple source systems/ devices, HINs/QHINs
		Multiple portals and/or repeated "pulls" to gather "current" view

Reconciled Lists		
Current and Historical • Medications • Problems, conditions • Allergies, intolerances • Immunizations • Immunization forecast • Procedures and interventions	Patient data in one place. In real-time, patient can: • Monitor data and actions • Curate lists • Be prompted to take actions • Complete and document actions taken	Patient data in multiple disparate fragments across multiple source systems/ devices, HINs/QHINs Multiple portals and/or repeated "pulls" to gather "current" view

ASTP/ONC seeks to explain its QHIN plans by noting the Cures Act requirement "to avoid the disruption of existing exchanges between participants of health information networks." Section 4003 of the Cures Act (amending Section 3001(c)(9) of the Public Health Service Act) does indeed require ONC to preserve capabilities for exchanging health information among at least some existing networks. It is to do this in part by establishing what has become TEFCA.

But there is no requirement to preserve exchange capability among *all* existing networks. And no requirement to cobble together the new networks that are QHINs.

Specifically, ASTP/ONC is not required to preserve any particular network in TEFCA's architecture. It is sufficient in TEFCA *to preserve health data exchange that includes interchange of data with EMR (Electronic Medical Record) networks in hospitals and physician offices.* That is enough to satisfy the statute.

Besides being a statutory truism, this is important as a practical matter; for many existing health exchange networks are only marginally functional. Those that have managed to collect individual patient data and maintain it in centralized records may at some point in the future be in position to enhance their capabilities to become Cures Act Repositories.

How can ASTP/ONC most efficiently fulfill the Cures Act's requirement "to avoid the disruption of existing exchanges between participants of health information networks?" Answer: By replacing point-to-point fax exchanges among EMR networks in hospitals and physician offices with FHIR-based, trusted, point-to-point network messages. Establishing directories of authenticated FHIR endpoints for such exchanges will include mechanisms for identity proofing, credentialing, and authorization of senders and recipients. This architecture is efficient, avoiding unnecessary complexity, security vulnerabilities, and costs.

To summarize: QHINs are unnecessary for any of TEFCA purposes as defined by the Cures Act. Indeed, inserting QHINs as unnecessary intermediaries between message endpoints adds useless cost and complexity while achieving no discernable function. In the process, inserting superfluous intermediary nodes and systems gratuitously multiplies attack surfaces and related security risks.

It is apparent that ASTP/ONC's QTF architecture is based on mis-reading the Cures Act's policy mandates and technical specifications. Inserting QHINs in digital health data exchanges is centered around medical institutions. It fails to elevate patients' powers to use and control their health information through a data architecture utilizing Cures Act repositories.

Overview of Enhanced Data Flows from a Cures Act Repository-Based Architecture

Each consumer has the right under HIPAA to designate a representative, including an online entity such as a Cures Act Repository/Health Data Bank, to receive, in digital form, a copy of their institutional (hospital or medical office) health record. If the patient so requests via a HIPAA authorization, this can include a copy of each subsequent provider encounter report. These records, presumably henceforth in FHIR format, can be aggregated, normalized, and stored securely in a Cures Act Repository/Health Data Bank account. For this purpose the Cures Act specifies:

> *The Secretary shall use existing authorities [...] with the goal of offering patients access to their electronic health information in a single, longitudinal format that is easy to understand, secure, and may be updated automatically.*

Parents can make this choice of a Cures Act Repository for their children. Adults can

choose a Cures Act Repository in the marketplace, and change their chosen repository at any time and for any reason.

Each organization offering Cures Act Repository/Health Data Bank account services must attest that it adheres to key principles of data privacy and security best practices. HRBA has long advocated for federal legislation to establish a nationwide regulatory structure to promote the development of Cures Act Repository/Health Data Banks, and to prevent bad actors from entering the HDB industry. In the absence of federal legislation, a health data bank that fails to adopt and adhere to industry-wide voluntary codes of conduct – and, frankly, that is operated to scam consumers – will likely be subject to investigation and enforcement action by state and local law enforcement agencies, by various law enforcement agencies, and by the Federal Trade Commission or HHS Office of Civil Rights. Intentional violations will offer prospects for criminal prosecution.

Congress's passing legislation to establish a nationwide regulatory structure now is highly desirable. Consumer and clinician trust will be essential to the emergence of Cures Act Repository/Health Data Banks as a structural feature of the nation's health data exchange architecture.

Whenever a patient receives a healthcare-related service from a hospital, medical office, or other provider, a record of that encounter will (because of the patient's HIPAA authorization and request) be sent electronically and in FHIR format to the account in their designated Cures Act Repository/Health Data Bank (unless the patient specifies otherwise).

When seeing a new provider, the patient can send the provider all or excerpted parts of their problem-oriented and normalized, aggregated record from their Cures Act Repository/ account. Alternatively, the patient can furnish the provider with access to all or a specified subset of their Cures Act Repository account. Either way, this system design enables the patient to supply the provider with essential information in useful formats, and promotes both provider convenience and comprehensive care.

Because the patient's lifetime record is available in one secure HDB account, in standardized FHIR format and normalized, online services offered by a Cures Act Repository or by external consulting services (some presumably by way of secure APIs) can furnish clinical guidance, help patients navigate care, or facilitate participation of patients in clinical research. They can obtain data from a patient's Cures Act Repository rather than having to query the patient or request access to various healthcare organizations' data systems where the patient has received care. A patient can also grant access to some or all of their Cures Act Repository lifetime record to family or other personal caregivers, to allow them to participate in the patient's care.

Subsequently, when healthcare providers use their EHR systems to access data available from a patient's aggregated and normalized Cures Act Repository account, the record (or a designated portion of it) will already be in a form that allows quick, accurate identification of elements relevant to the current patient encounter. It will be searchable. The patients' Cures Act Repository services or third-party curation services of the patient's choice can enhance the record to increase its usability for effective care and for the patient's understanding.

Fast Healthcare Interoperability Resources
Enable Cures Act Repository Architecture

With the advent of Fast Healthcare Interoperability Resources, the technology at last is available to extract medical records conveniently and affordably from electronic medical record (EMR) systems siloed in hospitals and physicians' offices. Standardized FHIR can make the process of extraction from the variety of EHR systems routine, efficient, and reliable. The resulting Cures Act records can then be moved into Cures Act Repository/Health Data Banks.

The architecture we recommend as a principal objective achieves dual interrelated objectives: first, extracting and then moving medical records efficiently; second, doing so in a way that encourages development of lifetime, problem-oriented Cures Act records stored in Cures Act Repositories under patients' control and ownership.

Institutions will retain their own medical records to document their processes with respect to each patient. This is a consequence of legal requirements in all states as well as for many federal purposes; and it is a practical necessity for operational purposes. Some institutions may create segregated or partitioned areas in their EHR systems to allow clinicians' viewing of imported Cures Act Records but without adding those imported records to the institution's own patient records. This may be done to preserve an institution's control over the data integrity of its internally generated records while allowing its clinicians to benefit from information in a patient's problem-oriented, up-to-data, searchable Cures Act Record.

Cures Act Repositories will be software-driven service centers for patients. Patients will use them to hold consolidated health record accounts they own, control, and use. The repositories will be somewhat akin to financial institutions. They will offer software systems to aggregate medical records from different providers into "lifetime" or "longitudinal," problem-oriented health record accounts. Patients will use their aggregated medical records to facilitate interactions with the health system writ large. Patients will, for example, be able to use repository software to direct their medical record data or portions of it to providers caring for them or to researchers whose work they want to support.

We expect Cures Act repositories to be basic structural components of a newly capable national health data exchange system centered around patients. (As we have said above, the most efficient way to organize medical data is to integrate it around each patient. The proposition is self-justifying.)

Networks to move patient data into and out of Cures Act Repository accounts – for example, to points of care in hospitals or physician offices, whether in routine or emergency circumstances – will function by using directories of authenticated FHIR endpoints for point-to-point exchanges. Patients, patient repository accounts, hospitals, physicians and physician offices, and pharmacies are examples of network destinations that will be authenticated and credentialed as FHIR endpoints.

FHIR is undergoing a continuing process of expansion and standardization. Its reach is being extended into more detailed areas of general medical practice and into medical specialties. This work is being accomplished by Health Level 7 (HL7), a standards development organization (*see* https://confluence.hl7.org/display/HL7/Understanding+the+Standards+Process; *see also* https://www.healthit.gov/topic/standards-technology/standards/fhir). HL7's ongoing

development of FHIR could be accelerated significantly via federal funding to implement existing Cures Act goals and engineering specifications.

FHIR is essentially becoming *the* national health data exchange standard. Having such a standard is a long-sought technology goal. FHIR is what enables engineers today to design data extraction and exchange architecture organized around the patient. As we have pointed out, this is in contrast to continuing to store patient records only in data systems scattered in various hospitals and physician offices.

A patient-centered data architecture will significantly improve information flows throughout medical care and research institutions. It will offer diagnostic and treatment efficiencies to clinicians, while easing their record-keeping burdens. It will simplify design of institutional data administration, and improve data exchange throughout and among health care institutions. It will improve care and greatly accelerate progress in medical research.

A FHIR-based Cures Act Repository architecture, with health data integrated around patients, represents a fundamental shift from institutionally organized legacy systems and architectures. The legacy systems are incapable of efficiently organizing and exchanging digital health data on a state, regional, or national basis.

The repositories can be private-sector or government-sponsored.

Patients who opt not to store their aggregated medical records in Cures Act Repositories will still enjoy the advantages of this streamlined system. Clinicians at points of care can make secure point-to-point queries to various providers that patients identify. That way they can obtain records from those sources via credentialed point-to-point messaging.

What does this straightforward system design based on the Cures Act offer? The architecture, based on patients' lifetime health records, exchangeable via digital point-to-point messaging, is relatively secure (compared to current security architectures) and simplified. It is operationally effective and cost-effective for nationwide use.

Of special significance: A Cures Act Repository architecture is a ready digital replacement for facsimile exchange of health data. (That is so even though secure messaging is slowly replacing fax.) This should be a central CMS and ASTP/ONC goal, because fax today, obsolete though it may seem, remains the de facto method for moving medical records in the U.S.

Enormous benefits in care, research, health equity, public health, and cost control will flow from this conversion.

This all contrasts with the experience of some large employers that sought to slow escalating workforce healthcare and health insurance costs. They hoped to do this by using medical record system consolidation to improve the health of their employees and their families (and thus reduce overall health insurance costs). In other words, major employers hoped to develop comprehensive medical record systems that members of their workforces would use – the assumption being that such use would result in improved health. In almost all cases, those efforts failed, some spectacularly and with great attention from the press.

The cause underlying these failures was not a clash of corporate cultures or similar public relations explanations. Rather, it was that employers' experimental data systems had no ability, at scale, to extract health records from scattered institutional EHR data bases in hospitals and medical offices. The proposed systems lacked the capacity to move data out of those institutional EHR systems and into employee health record accounts. Without that capability, the experimental employer systems could not aggregate and compile lifetime patient data, in useful form, that their employees would trust and use.

The Cures Act architecture we propose addresses these problems directly, and solves them.

Conclusion

CMS asks questions that bring to the fore opportunities to use FHIR technology to revolutionize health information data flows. The key is to use technology to support integrating health data round each patient. To do this requires a system architecture with Cures Act Repository/Health Data Banks as an essential structural element of the health information infrastructure. Cures Act Repository/Health Data Banks will serve as the vehicle to receive copies of health data from multiple sources, aggregate and normalize it, and hold the constantly updated, lifetime records in secure accounts that patients own, control and use.

Respectfully submitted,
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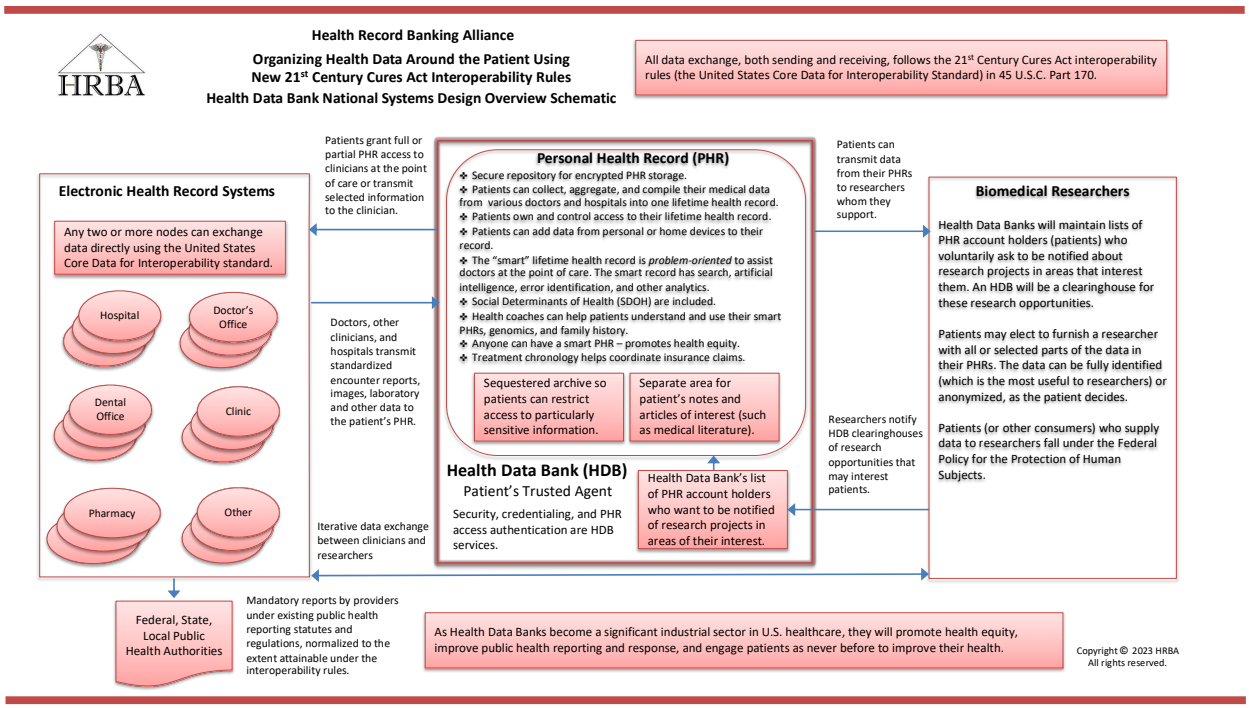
Appendix 1

Cures Act Health Data Bank Personal Health Record Schematic and Explanatory Text



Note that patient accounts in this Cures Act Repository/Health Data Bank schematic have unique patient identifiers. Those account numbers are analogous to depositors' checking and savings account numbers in financial institutions. For patients with Cures Act Repository/HDB accounts, this feature significantly ameliorates, and may largely eliminate for patients using HDB accounts, patient matching and attendant issues that are of great concern in other health data architectures.

The business models of many Cures Act HDBs will likely include separately encrypting the medical and other health records for each account holder. The system reason for separately encrypting each account is to complicate and largely prevent successful hacking of HDB accounts by hostile actors. Each account holder will be entrusted with the key to their encrypted account. The HDB will have procedures so that recovery of misplaced, lost, or corrupted keys will be possible for those account holders who desire that feature.

On the following page you will find additional explanatory text describing features and operation of this Cures Act Repository/HDB schematic.

Health Record Banking Alliance

Organizing Health Data Around the Patient Using New 21st Century Cures Act Interoperability Rules

Health Data Bank National Systems Design Overview

A **Health Data Bank** (**HDB**, also called a Health Record Bank) is an **integrated patient information services institution**. As a **trusted agent**, it offers a **secure repository** for each individual to collect and compile their **"interoperable"** digital health information in a **smart Personal Health Record (PHR).** Individuals own and control their Personal Health Records, as in a bank checking account. With these **new information flows**, consumers will:

- exchange medical records and other health data in their Personal Health Records conveniently with doctors' offices and hospitals for better, faster care; improve patient safety; and reduce information burden on physicians by supplying an aggregated, lifetime, searchable medical record for easy and immediate reference.
- control Personal Health Record access for doctors and hospitals; family, friends, and health coaches; medical researchers; members of the press; and others as they wish.
- use their Personal Health Records to help manage their health and healthcare, and to help shop for doctors, hospitals, and health insurance.
- view their Personal Health Records on smartphones, tablets, and other computers.

Health Data Banks and Efficiency: Integrating health information around each patient via HDBs is the most efficient way to aggregate and use **"interoperable"** health data under 21st Century Cures Act regulations. It is far more efficient and useful than a collection of "apps."

HRBA's Education and Policy Advocacy: HRBA advocates government policies promoting Health Data Banks as **a major new structural sector in U.S. health care**. This systems design includes a **national regulatory framework for Health Data Banks**.

Health Data Banks and Health Equity: Health Data Banks will promote **health equity** because **everyone can have a Personal Health Record**.

Health Data Banks as Medical Research Clearinghouses: Medical researchers cannot get enough patient data to make fast or sufficient progress. HDBs can be clearinghouses between patients and researchers. Patients can **voluntarily** list themselves with their HDBs to be informed of research projects they are interested in, and to which they want to **contribute or sell their data**. This also is a path to developing **national federated diagnostic and research databases** while respecting **patients' privacy rights** (because patients are in control). Better research will improve treatment for acute, chronic, and orphan diseases.

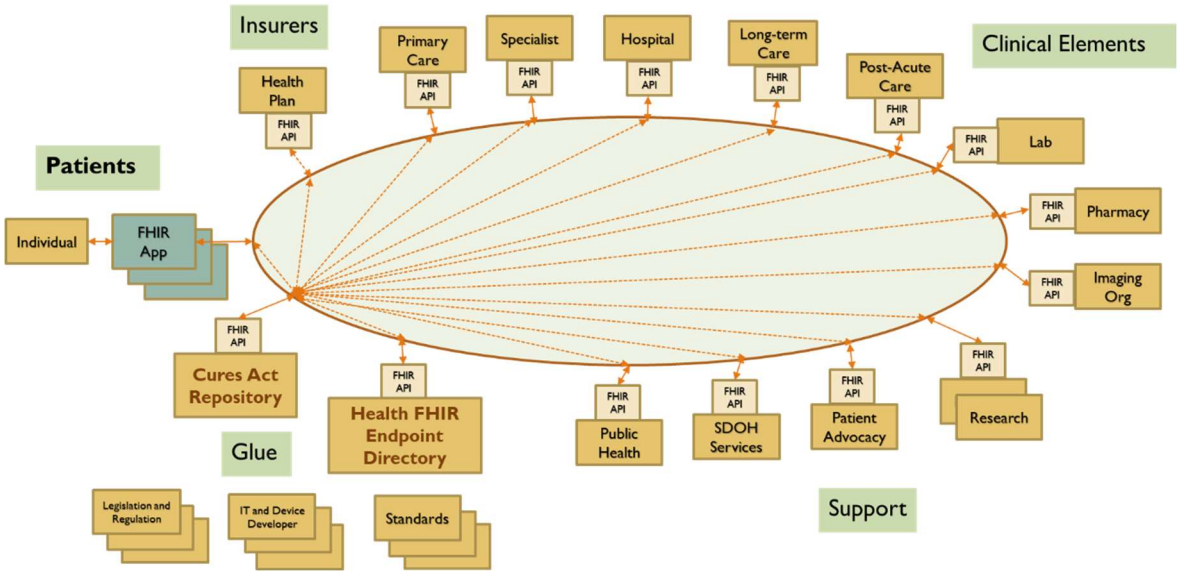
Health Data Banks, Security, and Patient Matching: Security, credentialing, and patient authentication and efficient matching are systems design features of HDBs

Advanced Features of Smart Personal Health Records: Systems design features such as artificial intelligence (AI) and search capabilities, robust family history, and genomic analytics will deliver **problem-oriented data and analysis** to **mesh** with clinicians' Electronic Health Record (EHR) systems **at the point of care**. Availability of this aggregated **reference record** will reduce burdens on clinicians while improving diagnosis, treatment, and patient outcomes.

Comments of
The Health Record Banking Alliance

Centers for Medicare and Medicaid Services
Request for Information – Health Technology Ecosystem
Reference File Code CMS–0042–NC

Appendix 2

Cures Act Repository/Health Data Bank-Based
Ecosystem Schematic
Based on Standardized FHIR
For Health Data Storage and Exchange

Note: Trust Over IP provides the tools to handle the encryption/decryption, messaging, and key management in this storage and exchange architecture.